
Fourier Transform

2D Fourier Transform

The formula for the **continuous 2D Fourier Transform (FT)** is:

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy,$$

where:

- $f(x, y)$: The spatial domain function (e.g., an image or signal).
- $F(u, v)$: The frequency domain representation of $f(x, y)$.
- u and v : Frequency variables corresponding to the x - and y -axes, respectively.
- $e^{-j2\pi(ux+vy)}$: The complex exponential kernel.

2D Inverse Fourier Transform

To recover $f(x, y)$ from its Fourier Transform $F(u, v)$, the inverse 2D Fourier Transform is given by:

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv.$$

2D Discrete Fourier Transform

- For $M \times N$ matrix, forward and inverse fourier transforms can be written

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \exp \left[-2\pi i \left(\frac{xu}{M} + \frac{yv}{N} \right) \right].$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \exp \left[2\pi i \left(\frac{xu}{M} + \frac{yv}{N} \right) \right].$$

2D Discrete Fourier Transform

Because the 2-D DFT is complex in general, it can be expressed in polar form:

$$F(u, v) = |F(u, v)|e^{j\phi(u, v)}$$

where the magnitude or Fourier (or frequency) spectrum

$$|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2}$$

phase angle:

$$\phi(u, v) = \arctan\left[\frac{I(u, v)}{R(u, v)}\right]$$

Finally, the power spectrum is defined as

$$\begin{aligned} P(u, v) &= |F(u, v)|^2 \\ &= R^2(u, v) + I^2(u, v) \end{aligned}$$

2D Discrete Fourier Transform

Given the signal $f(x) = [1, 2, 3, 4]$, compute its Discrete Fourier Transform (DFT) using the formula:

$$F(k) = \sum_{x=0}^{N-1} f(x) \cdot e^{-j \frac{2\pi kx}{N}}, \quad k = 0, 1, \dots, N-1.$$

Here, $N = 4$.

2D Discrete Fourier Transform

1. For $k = 0$

$$F(0) = 1 + 2 + 3 + 4 = 10$$

2. For $k = 1$

Compute each term:

- $e^{-j2\pi(1)(0)/4} = 1$
- $e^{-j2\pi(1)(1)/4} = e^{-j\pi/2} = -j$
- $e^{-j2\pi(1)(2)/4} = e^{-j\pi} = -1$
- $e^{-j2\pi(1)(3)/4} = e^{-j3\pi/2} = j$

Now plug into the sum:

$$\begin{aligned} F(1) &= 1(1) + 2(-j) + 3(-1) + 4(j) \\ &= 1 - 3 + (-2j + 4j) \\ &= -2 + 2j \end{aligned}$$

$F = [10, -2 + 2j, -2, -2 - 2j]$

3. For $k = 2$

- $e^{-j2\pi(2)(0)/4} = 1$
- $e^{-j2\pi(2)(1)/4} = e^{-j\pi} = -1$
- $e^{-j2\pi(2)(2)/4} = e^{-j2\pi} = 1$
- $e^{-j2\pi(2)(3)/4} = e^{-j3\pi} = -1$

Now compute:

$$\begin{aligned} F(2) &= 1(1) + 2(-1) + 3(1) + 4(-1) \\ &= 1 - 2 + 3 - 4 \\ &= -2 \end{aligned}$$

4. For $k = 3$

Use symmetry:

$F(3)$ is the complex conjugate of $F(1)$.

$$F(3) = -2 - 2j$$

Properties of 2D DFT

Problem:

Given the 2D image:

$$f(x, y) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

compute its 2D DFT using the formula:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)},$$

where $M = N = 2$.

Properties of 2D DFT

a) $F(0, 0)$

Here $u = 0, v = 0$:

$$ux + vy = 0 \Rightarrow (-1)^{ux+vy} = 1$$

$$\begin{aligned} F(0, 0) &= \sum_{x=0}^1 \sum_{y=0}^1 f(x, y) \cdot 1 \\ &= 1 + 2 + 3 + 4 = 10 \end{aligned}$$

b) $F(0, 1)$

Here $u = 0, v = 1$:

$$ux + vy = y \Rightarrow (-1)^{ux+vy} = (-1)^y$$

$$\begin{aligned} F(0, 1) &= f(0, 0)(-1)^0 + f(0, 1)(-1)^1 + f(1, 0)(-1)^0 + f(1, 1)(-1)^1 \\ &= 1(1) + 2(-1) + 3(1) + 4(-1) \\ &= 1 - 2 + 3 - 4 \\ &= -2 \end{aligned}$$

c) $F(1, 0)$

Here $u = 1, v = 0$:

$$ux + vy = x \Rightarrow (-1)^{ux+vy} = (-1)^x$$

$$\begin{aligned} F(1, 0) &= f(0, 0)(-1)^0 + f(0, 1)(-1)^0 + f(1, 0)(-1)^1 + f(1, 1)(-1)^1 \\ &= 1(1) + 2(1) + 3(-1) + 4(-1) \\ &= 1 + 2 - 3 - 4 \\ &= -4 \end{aligned}$$

d) $F(1, 1)$

Here $u = 1, v = 1$:

$$ux + vy = x + y \Rightarrow (-1)^{ux+vy} = (-1)^{x+y}$$

$$\begin{aligned} F(1, 1) &= f(0, 0)(-1)^{0+0} + f(0, 1)(-1)^{0+1} + f(1, 0)(-1)^{1+0} + f(1, 1)(-1)^{1+1} \\ &= 1(1) + 2(-1) + 3(-1) + 4(1) \\ &= 1 - 2 - 3 + 4 \\ &= 0 \end{aligned}$$

$$F = \begin{bmatrix} 10 & -2 \\ -4 & 0 \end{bmatrix}$$

Properties of 2D DFT

Problem:

Given the Fourier coefficients:

$$F(k) = [10, -2 + 2j, -2, -2 - 2j],$$

find the original signal $f(x)$ using the inverse DFT formula:

$$f(x) = \frac{1}{N} \sum_{k=0}^{N-1} F(k) \cdot e^{j \frac{2\pi kx}{N}}, \quad x = 0, 1, \dots, N-1.$$

Here, $N = 4$.

Properties of 2D DFT

Problem:

Given the 2D Fourier transform coefficients:

$$F(u, v) = \begin{bmatrix} 10 & -2 \\ -4 & 0 \end{bmatrix},$$

find the original 2D image $f(x, y)$ using the inverse 2D DFT formula:

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \cdot e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)},$$

where $M = N = 2$.

Separability

Notice that Fourier transform “filter elements” can be expressed as products

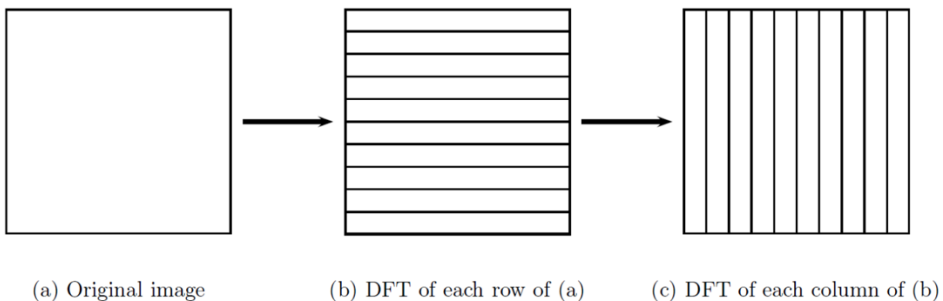
$$\exp \left[2\pi i \left(\frac{xu}{M} + \frac{yv}{N} \right) \right] = \underbrace{\exp \left[2\pi i \frac{xu}{M} \right]}_{\text{2D DFT}} \underbrace{\exp \left[2\pi i \frac{yv}{N} \right]}_{\text{1D DFT (row) 1D DFT (column)}}$$

Formula above can be broken down into simpler formulas for 1D DFT

$$F(u) = \sum_{x=0}^{M-1} f(x) \exp \left[-2\pi i \frac{xu}{M} \right],$$
$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) \exp \left[2\pi i \frac{xu}{M} \right]$$

Separability

- Using their separability property, can use 1D DFTs to calculate rows then columns of 2D Fourier Transform



Separability

- Can use separability to implement 2D DFT as sequence of 1D DFTs on rows and columns

```
1: SEPARABLE 2D-DFT ( $g$ )           ▷  $g(u, v) \in \mathbb{C}, 0 \leq u < M, 0 \leq v < N$ 
2:   for  $v \leftarrow 0 \dots N-1$  do
3:     Let  $g(\cdot, v)$  be the  $v$ th row vector of  $g$ :
       Replace  $g(\cdot, v)$  by  $\text{DFT}(g(\cdot, v))$ 
4:   for  $u \leftarrow 0 \dots M-1$  do
5:     Let  $g(u, \cdot)$  be the  $u$ th column vector of  $g$ :
       Replace  $g(u, \cdot)$  by  $\text{DFT}(g(u, \cdot))$ 
   Remark:  $g(u, v) \equiv G(u, v) \in \mathbb{C}$  now contains the discrete 2D spec-
   trum.
6:   return  $g$ 
```

Suppose we have a 2×2 signal matrix $f(x, y)$:

$$f(x, y) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Separability

Problem:

Given the 2D image:

$$f(x, y) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

verify that the 2D DFT can be computed by:

1. Applying the 1D DFT to each row.
2. Applying the 1D DFT to each column of the result.

Problem:

Compute the 2D DFT for the image:

$$f(x, y) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

by applying separability (first rows, then columns).

Separability

1. Apply 1D DFT to Rows:

Compute the DFT for each row:

- First row: $[1, 2]$

$$F_{\text{row}}(0) = 1 + 2 = 3, \quad F_{\text{row}}(1) = 1 - 2 = -1.$$

Result: $[3, -1]$.

- Second row: $[3, 4]$

$$F_{\text{row}}(0) = 3 + 4 = 7, \quad F_{\text{row}}(1) = 3 - 4 = -1.$$

Result: $[7, -1]$.

After row-wise DFT:

$$F_{\text{rows}} = \begin{bmatrix} 3 & -1 \\ 7 & -1 \end{bmatrix}.$$

2. Apply 1D DFT to Columns:

Now, compute the DFT for each column of F_{rows} :

- First column: $[3, 7]$

$$F_{\text{col}}(0) = 3 + 7 = 10, \quad F_{\text{col}}(1) = 3 - 7 = -4.$$

Result: $[10, -4]$.

- Second column: $[-1, -1]$

$$F_{\text{col}}(0) = -1 - 1 = -2, \quad F_{\text{col}}(1) = -1 - (-1) = 0.$$

Result: $[-2, 0]$.

After column-wise DFT:

$$F(u, v) = \begin{bmatrix} 10 & -2 \\ -4 & 0 \end{bmatrix}.$$

Scaling

$$f(ax, by) \Leftrightarrow \frac{1}{ab} F\left(\frac{u}{a}, \frac{v}{b}\right)$$

Scaling

Problem:

The Fourier transform of $f(x) = e^{-x^2}$ is given as:

$$F(u) = \sqrt{\pi} e^{-\pi^2 u^2}.$$

Find the Fourier transform of $f(2x)$.

Problem:

For a 2D signal $f(x, y) = e^{-(x^2+y^2)}$, the Fourier transform is:

$$F(u, v) = \pi e^{-\frac{(u^2+v^2)}{4}}.$$

Find the Fourier transform of $f(2x, 3y)$.

Problem:

Given a Fourier transform $F(u) = \text{sinc}(u)$, find the signal in the spatial domain when $F(u)$ is scaled as $F(3u)$.

Convolution

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$$

$$f(x, y) \star h(x, y) \Leftrightarrow F(u, v) H(u, v)$$

$$f(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$$

Convolution

DFT provides alternate method to do **convolution** of image M with spatial filter S

1. Pad S to make it same size as M , yielding S'
2. Form DFTs of both M and S'
3. Multiply M and S' element by element

$$F(M) \times F(S')$$

Take inverse transform of result

$$F^{-1}(F(M) \times F(S'))$$

DC Component

Recall that

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

The value $F(0,0)$ of the DFT is called the **dc coefficient**

If we put $u = v = 0$, then

$$\begin{aligned} F(0, 0) &= MN \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \\ &= MN \bar{f}(x, y) \end{aligned}$$

Essentially $F(0,0)$ is the sum of all terms in the original matrix

DC Component

Consider a 2D signal $f(x, y)$ with size $M \times N$ and the following values:

$$f(x, y) = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Task: Find the DC component of this signal.

Consider the signal $f(x, y)$:

$$f(x, y) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Task: Find the DC component of this signal.

DC Component

Problem:

Given the 1D signal:

$$f(x) = [1, 2, 3, 4]$$

Tasks:

1. Compute the **1D Discrete Fourier Transform (DFT)** of $f(x)$.
2. Find the **DC component** of the signal, which is the value of the Fourier transform at $u = 0$.

Problem:

Given the 2D signal:

$$f(x, y) = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Tasks:

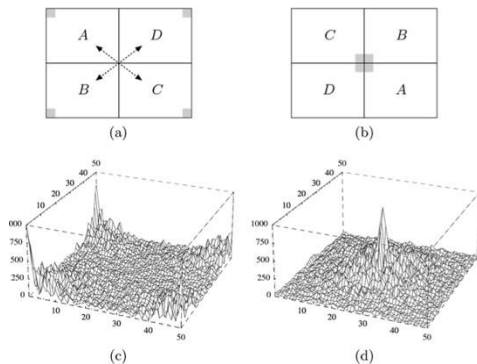
1. Compute the **2D Discrete Fourier Transform (DFT)** of $f(x, y)$.
2. Find the **DC component** of the 2D signal, which is the value of the Fourier transform at $u = 0$ and $v = 0$.

Centering DFT Spectrum

$F(0,0)$ at top left corner

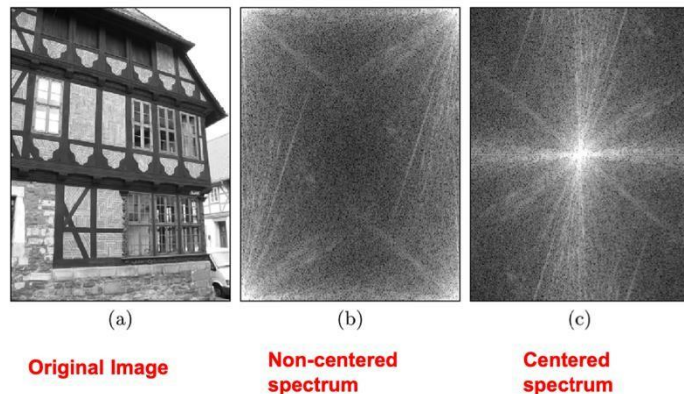
For display, convenient to have DC component in center

Just swap four quadrants of Fourier transform



Swap 4 quadrants to
center DC component

DFT spectrum after
centering



Displaying Transforms

- As elements are complex numbers, we can view magnitudes $|F(u, v)|$ directly.
- Displaying magnitudes of Fourier transforms called **spectrum** of the transform.
- **Problem:** DC component much larger than other values.
Displays white dot in middle surrounded by black.
- So stretch transform values by displaying log of transform .

$$\log(1 + |F(u, v)|)$$

The Basics of Filtering in the Frequency Domain

Filtering in the frequency domain consists of modifying the Fourier transform of an image and then computing the inverse transform to obtain the processed result. Thus, given a digital image, (x, y) , of size $M \times N$, the basic filtering equation in which we are interested has the form:

$$g(x, y) = \mathfrak{F}^{-1}[H(u, v)F(u, v)]$$

Where $\mathfrak{F}^{-1}$ is the IDFT, (u, v) is the DFT of the input image, (x, y) , $H(u, v)$ is a filter function (also called simply the filter) and $g(x, y)$ is the filtered (output) image. Functions F , H , and g are arrays of size the same as the input image

The Basics of Filtering in the Frequency Domain

The material in the previous two sections can be summarized as follows:

1. Given an input image (x, y) of size obtain the padding parameters P and Q . Typically, we select $P = 2M$ and $Q = 2N$
2. Form a padded image, $f_p(x, y)$ of size $P \times Q$ by appending the necessary number of zeros to $f_p(x, y)$
3. Multiply $f_p(x, y)$ by $(-1)^{x+y}$ to center its transform
4. Compute the DFT $F(u, v)$, of the image from step 3.
5. Generate a real, symmetric filter function, $H(u, v)$, of size $P \times Q$ with center at coordinates $(P/2, Q/2)$ Form the product $G(u, v) = H(u, v)F(u, v)$ using array multiplication; that is $G(i, k) = H(i, k)F(i, k)$.
6. Obtain the processed image:

$$g_p(x, y) = \{\text{real}[\mathfrak{F}^{-1}[G(u, v)]]\}(-1)^{x+y}$$

where the real part is selected in order to ignore parasitic complex components resulting from computational inaccuracies, and the subscript p indicates that we are dealing with padded arrays.

7. Obtain the final processed result, $g(x, y)$, by extracting the $M \times N$ region from the top, left quadrant of $g_p(x, y)$

The Basics of Filtering in the Frequency Domain

As noted earlier, low frequencies in the transform are related to slowly varying intensity components in an image, such as the walls of a room or a cloudless sky in an outdoor scene.

On the other hand, high frequencies are caused by sharp transitions in intensity, such as edges and noise

The Basics of Filtering in the Frequency Domain

1. Image Dominated by Low Frequencies

Characteristics of the image:

- Smooth, uniform areas with little intensity variation.
- Examples: clear skies, flat walls, or smooth backgrounds.

Fourier spectrum appearance:

- Brightness is concentrated at the center (low frequencies dominate).
- Peripheral areas (high frequencies) are dim or dark.

The Basics of Filtering in the Frequency Domain

2. Image Dominated by High Frequencies

Characteristics of the image:

- Contains sharp edges, fine details, or noise.
- Examples: textured surfaces, noisy images, or scenes with many fine patterns.

Fourier spectrum appearance:

- Brightness is concentrated at the center (low frequencies dominate).
- Peripheral areas (high frequencies) are dim or dark.

The Basics of Filtering in the Frequency Domain

3. Image Contains Both Low and High Frequencies

Characteristics of the image:

- A mix of smooth regions and detailed areas.
- Examples: landscapes with smooth skies and detailed objects, or portraits with textures and gradients.

Fourier spectrum appearance:

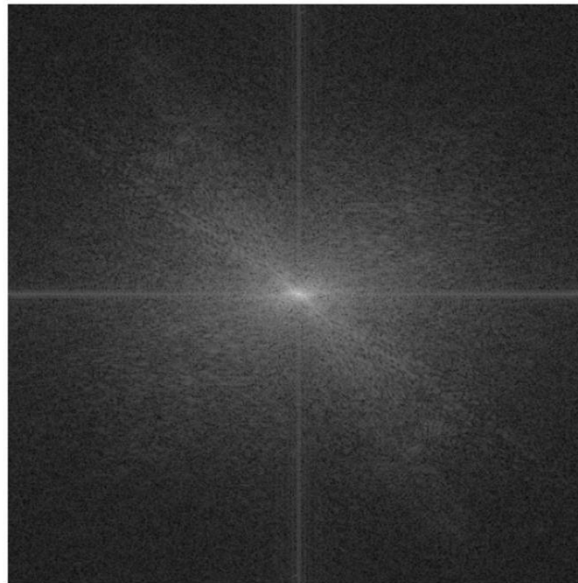
- The center is bright (low frequencies).
- Bright spots or energy in the outer areas represent high-frequency content.

The Basics of Filtering in the Frequency Domain

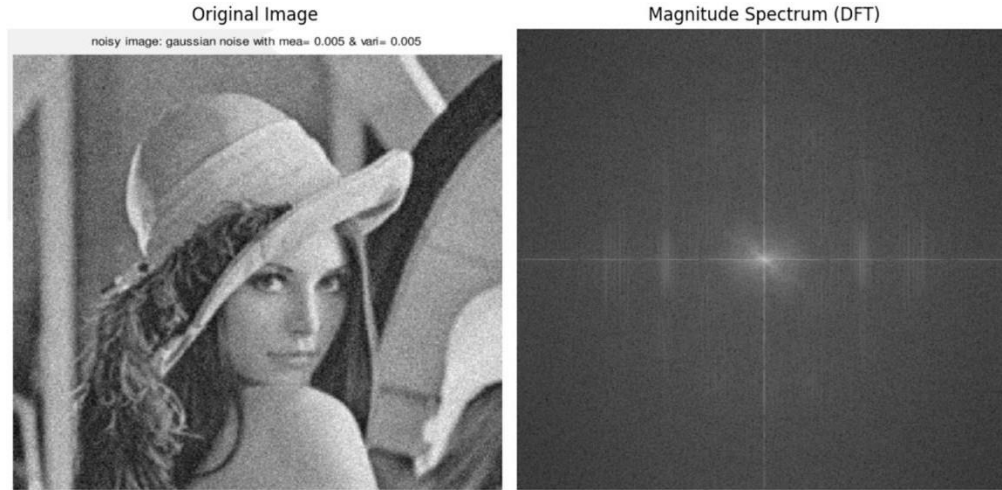
Original Image



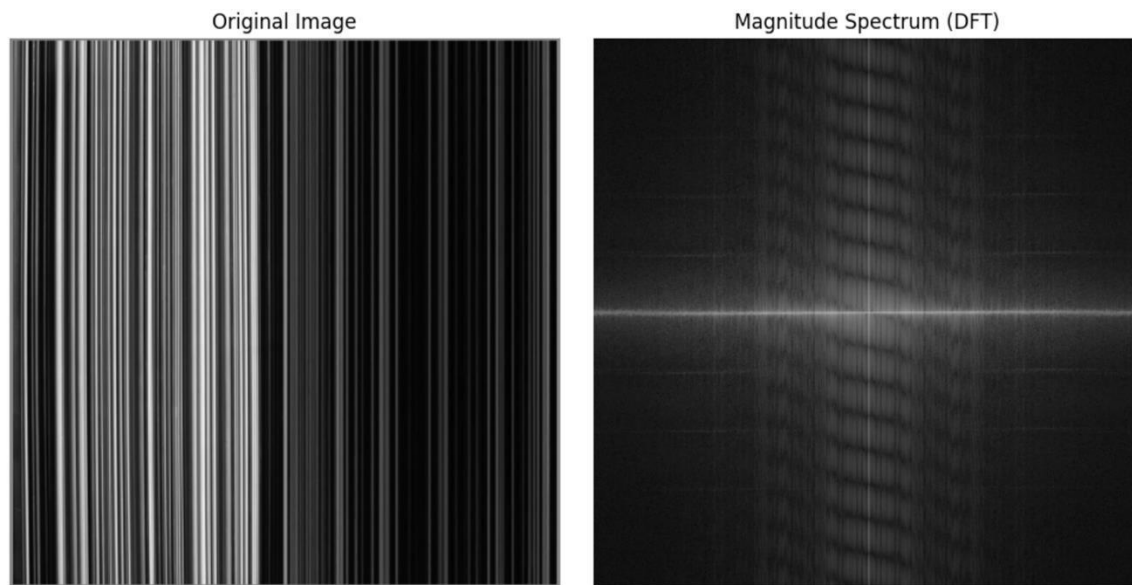
Magnitude Spectrum (DFT)



The Basics of Filtering in the Frequency Domain



The Basics of Filtering in the Frequency Domain



The Basics of Filtering in the Frequency Domain

4. Image with Periodic Patterns (Repetitive Structure)encies

Characteristics of the image:

- Contains repetitive or cyclic patterns.
- Examples: stripes, sinusoidal waves, or regularly repeating textures.

Fourier spectrum appearance:

- Distinct bright spots in the spectrum, corresponding to the periodic frequencies.
- These spots can occur both near the center and at the periphery, depending on the frequency of the pattern.

The Basics of Filtering in the Frequency Domain

a	b	c
d	e	f
g	h	

- (a) An $M \times N$ image, f .
- (b) Padded image f_p of size $P \times Q$.
- (c) Result of multiplying f_p by $(-1)^{x+y}$.
- (d) Spectrum of f_p .
- (e) Centered Gaussian lowpass filter, H , of size $P \times Q$.
- (f) Spectrum of the product HF_p .
- (g) g_p , the product of $(-1)^{x+y}$ and the real part of the IDFT of HF_p .
- (h) Final result, g , obtained by cropping the first M rows and N columns of g_p .

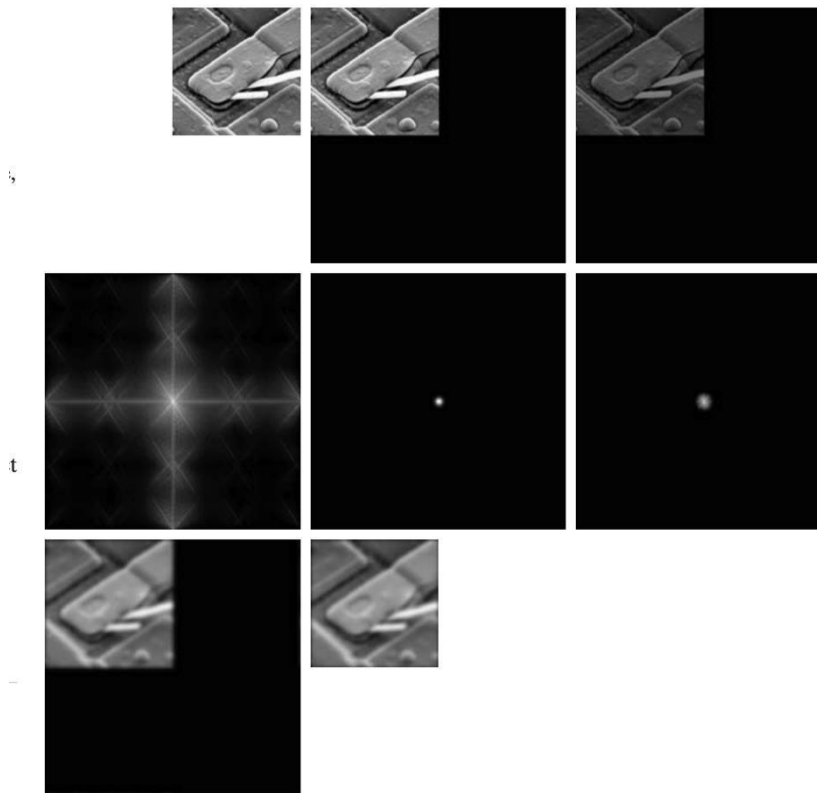


Image Smoothing Using Frequency Domain Filters

Ideal Lowpass Filters

A 2-D lowpass filter that passes without attenuation all frequencies within a circle of radius from the origin and “cuts off” all frequencies outside this circle is called an ideal lowpass filter (ILPF); it is specified by the function

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

Image Smoothing Using Frequency Domain Filters

Ideal Lowpass Filters

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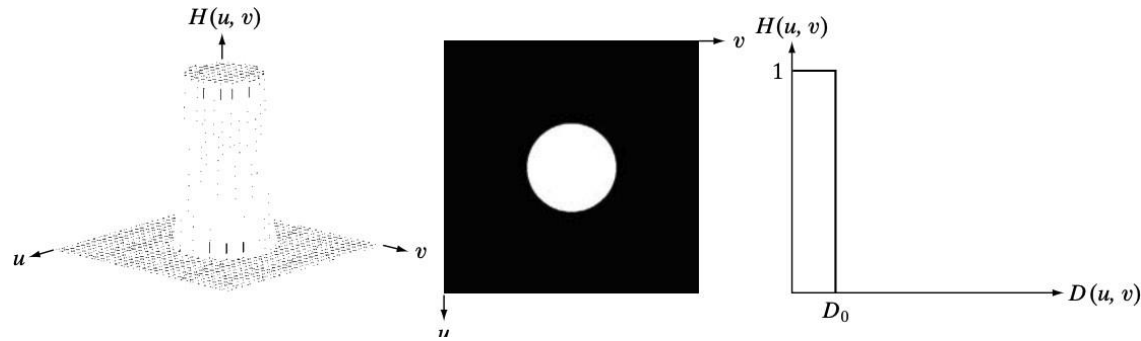
$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

Where D_0 is a positive constant and $D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the frequency rectangle; that is,

$$D(u, v) = [(u - P/2)^2 + (v - Q/2)^2]^{1/2}$$

where, as before, P and Q are the padded sizes

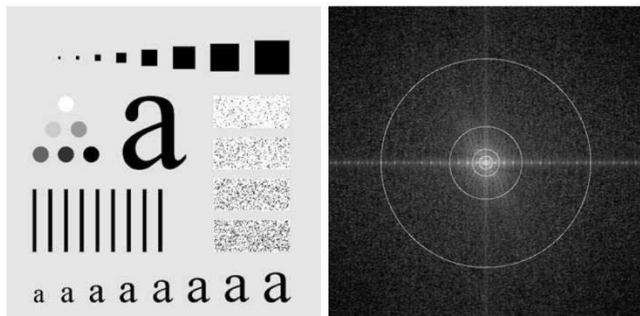
Image Smoothing Using Frequency Domain Filters



a b c

(a) Perspective plot of an ideal lowpass-filter transfer function. (b) Filter displayed as an image.
(c) Filter radial cross section.

Image Smoothing Using Frequency Domain Filters



a	b
c	d
e	f

(a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

